

EIELSON AFB, AK, USA

WMO: 702650

Lat: 64.683N

Lon: 147.083W

Elev: 547

StdP: 14.41

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26407

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.5	-37.0	-39.9	0.6	-32.0	-35.1	0.8	-28.4	17.9	29.3	15.2	21.6	1.4	170	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	81.0	60.7	77.4	60.0	74.3	58.6	63.8	76.5	61.8	73.3	60.2	70.6	6.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	76.6	66.2	57.2	71.2	64.3	55.4	66.6	63.3	29.2	76.7	27.8	73.4	26.7	70.9	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	13.6	11.5	DB	-44.1	86.9	6.3	3.2	-48.6	89.2	-52.3	91.1	-55.9	92.8	-60.4	95.2	(4)
(5)			WB	-37.2	67.0	6.9	2.3	-42.2	68.6	-46.2	70.0	-50.1	71.3	-55.1	72.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	-7.0	2.2	13.1	34.1	49.4	59.2	61.7	56.0	45.5	24.3	3.4	-5.6		
	DBStd	27.76	18.25	14.91	14.13	10.70	7.51	5.77	4.72	5.71	7.38	11.94	14.75	15.94		
	HDD50	8927	1766	1339	1145	479	100	2	0	11	167	798	1397	1723		
	HDD65	13494	2231	1759	1610	928	483	191	123	285	586	1263	1847	2188		
	CDD50	953	0	0	0	1	83	277	362	198	32	0	0	0		
	CDD65	41	0	0	0	0	0	16	19	6	0	0	0	0		
	CDH74	751	0	0	0	0	58	284	296	113	0	0	0	0		
	CDH80	125	0	0	0	0	4	55	45	21	0	0	0	0		
	WSAvg	3.9	2.3	2.6	3.9	5.3	5.6	5.1	4.5	4.1	4.3	3.4	2.7	2.5		
	PrecAvg	12.6	0.4	0.4	0.3	0.3	0.7	2.0	2.9	2.5	1.4	0.8	0.5	0.6		
(15) Precipitation	PrecMax	18.7	1.6	1.5	1.4	2.3	2.2	3.5	7.7	5.6	2.6	1.7	1.7	2.3		
	PrecMin	9.6	0.0	0.0	0.0	0.0	0.2	0.7	0.7	0.5	0.4	0.2	0.0	0.0		
	PrecStd	2.6	0.4	0.4	0.3	0.5	0.5	0.7	1.8	1.1	0.7	0.4	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	40.6	50.0	65.1	80.0	85.2	84.4	82.6	69.9	55.3	41.4	41.2		
		MCWB	31.4	33.7	39.6	47.8	56.3	61.5	63.5	63.6	54.6	43.5	34.4	34.5		
	2%	DB	32.1	34.4	44.3	59.8	74.9	81.2	80.7	77.0	66.0	49.2	32.8	33.0		
		MCWB	28.9	30.0	35.8	44.6	54.5	59.5	61.9	61.9	52.3	40.9	30.8	29.5		
	5%	DB	24.4	28.5	39.8	55.4	70.3	77.1	77.2	72.0	63.2	43.9	27.3	24.7		
		MCWB	23.3	26.3	32.8	42.4	52.1	59.0	61.2	58.9	50.9	38.3	25.9	23.4		
	10%	DB	17.6	23.1	35.5	52.1	66.1	73.4	74.0	68.1	59.8	39.2	23.3	17.7		
		MCWB	16.5	21.7	30.0	40.6	50.2	57.2	60.1	57.0	50.1	35.7	22.3	17.0		
	0.4%	WB	32.8	34.3	39.9	48.3	57.4	64.6	67.0	66.6	56.1	44.6	35.3	35.3		
		MCDB	36.2	39.8	48.6	63.4	75.4	80.5	77.3	78.9	65.4	53.4	40.4	40.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.7	30.5	36.2	45.6	55.7	62.0	64.2	62.8	53.7	41.7	30.9	29.8		
		MCDB	32.2	34.5	43.3	58.9	72.9	75.5	76.5	73.6	62.8	47.6	33.5	33.8		
	5%	WB	23.9	26.7	33.4	43.2	53.4	60.1	62.4	60.3	52.3	38.9	26.2	24.0		
		MCDB	25.5	28.9	39.7	54.8	67.6	73.3	74.1	68.6	60.7	43.6	27.5	25.4		
	10%	WB	17.3	21.9	30.1	41.0	51.1	58.4	60.9	58.5	50.7	35.9	22.6	17.3		
		MCDB	18.2	23.4	35.1	51.4	64.2	70.8	71.7	66.0	58.2	39.3	23.7	18.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	20.1	27.0	23.3	24.2	21.0	18.9	18.6	19.6	15.9	14.8	14.9		
		MCDBR	22.8	22.8	28.2	28.5	31.3	27.8	26.4	26.0	26.7	22.3	16.7	23.8		
		MCWBR	19.9	19.4	20.6	15.9	14.5	11.8	11.8	12.7	15.1	15.1	14.7	21.1		
		MCDBR	22.1	21.8	25.9	27.2	27.9	24.3	23.0	21.4	23.1	20.2	16.7	23.7		
	5% WB	MCWBR	19.2	18.5	18.8	15.5	13.0	10.9	11.5	10.9	13.7	13.9	14.6	20.9		
(40) Clear-Sky Solar Irradiance	taub		0.173	0.249	0.255	0.319	0.344	0.382	0.375	0.395	0.307	0.263	N/A	0.178		
	taud		1.949	2.202	2.300	2.261	2.429	2.332	2.380	2.310	2.550	2.462	N/A	1.941		
	Ebn at Noon		125	213	269	270	275	264	262	242	251	214	N/A	40		
	Edh at Noon		7	17	24	33	31	35	32	32	20	14	N/A	3		
(44) All-Sky Solar Radiation	RadAvg		53	273	841	1439	1653	1738	1512	1162	731	305	78	16		
	RadStd		4	22	55	91	123	144	105	113	85	34	10	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+1.55	N/A	N/A	N/A	N/A	N/A	-587	-637	N/A	N/A

Nomenclature: See separate page